

Qatar-1b

R-0621

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_240701 · created 2026-07-25T12:29:00+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460492.7519 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.054
Transit depth (Rp/Rs)^2	3.06 +/- 1.88 [%]
Orbital Inclination (inc)	81.57 +/- 2.91
Airmass coefficient 1 (a1)	1.17 +/- 0.17
Airmass coefficient 2 (a2)	-0.06 +/- 0.16
Scatter in the residuals of the lightcurve fit is	14.55 %
Best Comparison Star	#2 - [479, 259]
Optimal Aperture	1.61
Optimal Annulus	13.43
Transit Duration (day)	0.046 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.054 vs 0.1463 (deviation 0.5σ)
Duration fit vs published	0.046 d vs 0.0692 d (deviation 0.78σ)
Mid-time O–C	-6.5 min
Depth SNR	3.1
Residual scatter	14.55 %

Light curve

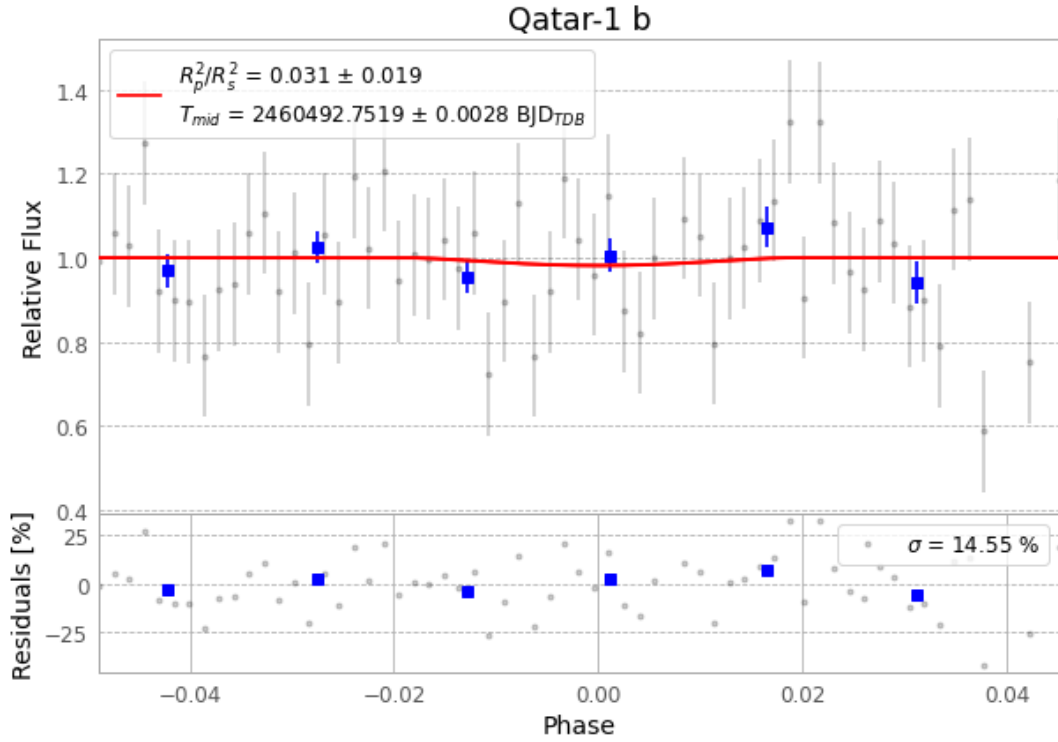


Figure 1 — FinalLightCurve_Qatar-1 b_2024-06-30.png (SHA-256 5e3ca9f38d41a61d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 178], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a8406361ac5ad435...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

73 FITS frames (48 MB), Qatar-1240701042115.FITS → Qatar-1240701075715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-240701.log (ef7d21791874...), FinalLightCurve_Qatar-1 b_2024-06-30.png (5e3ca9f38d41...), FinalLightCurve_Qatar-1 b_2024-06-30.csv (2a41a1a7aa5b...)

Compute resources

Wall time 230.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 12:29Z.